

Amazon Sales Data Analysis

End-to-End Data Analytics Portfolio Project | PostgreSQL & Power BI

1. Project Overview & Tech Stack

This comprehensive data analytics project focuses on cleaning, querying, and visualizing an Amazon sales dataset to extract actionable business insights.

PostgreSQL

Advanced SQL

Data Cleaning

Power BI Desktop

Data Visualization

2. Data Engineering & Cleaning (PostgreSQL)

The raw dataset contained non-numeric symbols (such as ₹, commas, and pipes) and inconsistencies that caused strict numeric casting failures (invalid input syntax errors).

- **Robust Transformation:** Utilized advanced PostgreSQL functions including `REGEXP_REPLACE`, `NULLIF`, and explicit casting (`::numeric`) to sanitize pricing, ratings, and rating counts.
- **Data Integrity:** Filtered out null and blank values to ensure analytical accuracy and prevent execution errors during aggregation.

3. Core Analytical Queries & Key Results

Four major business and exploratory queries were developed and executed:

- **Price Range Segmentation:** Segmented products into distinct pricing tiers to analyze market distribution:

Budget Products: < ₹500 | **Mid-Range Products:** ₹500 – ₹2000 | **Premium Products:** > ₹2000

- **Discount Analysis:** Computed exact monetary differences and percentage discounts between actual and discounted prices using precise rounding functions.
- **Category-Wise Performance:** Evaluated product counts, average ratings, and review distribution across multiple categories.
- **Popularity Metrics:** Identified top-reviewed and high-engagement products within the catalog.

4. Interactive Dashboard & Visualization (Power BI)

Integrated the sanitized PostgreSQL database (`ecommercer_analytics`) with Power BI Desktop via secure local connections (`localhost`) utilizing unencrypted connection handling.

- **Line & Stacked Column Chart:** Displayed category-wise Total Product Count via columns alongside Average Rating trends via data lines.
- **Price Distribution Donut Chart:** Visualized the market share across Budget, Mid-Range, and Premium pricing tiers.
- **Summary KPI Cards:** Highlighted high-level aggregated metrics including total product count and overall averages.